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Fostering Metacognitive Regulation Through Technology-Mediated Explain-to-Learn: A Design-Based Investigation of Microlearning and the Feynman Technique in Academic Speaking

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Abstract

This design-based study investigates how technology-mediated explain-to-learn activities foster metacognitive regulation as a mediating mechanism for academic speaking development. Grounded in the Cognitive Theory of Multimedia Learning (CTML) and self-regulated learning theory, the study examined the integration of microlearning and the Feynman technique among 96 Indonesian undergraduate students over eight weeks. Results revealed significant improvements in metacognitive regulation (Wilks' $\Lambda = .41$, $p < .001$, partial $\eta^2 = .59$) and academic speaking performance (Cohen's $d = 1.01$), with the strongest effects observed in planning and evaluation dimensions. Analysis of student perceptions indicated high perceived usefulness and cognitive engagement. The findings demonstrate that metacognitive regulation functions as a critical pathway linking explain-to-learn instructional design to speaking performance, offering evidence-based design principles for technology-enhanced language education. The study contributes to understanding how generative learning strategies embedded in microlearning environments can systematically activate regulatory processes essential for academic communication competence.

Keywords: metacognitive regulation; explain-to-learn; microlearning; Feynman technique; academic speaking; design-based research; generative learning

Introduction

The increasing integration of digital technologies into higher education has fundamentally reshaped how academic skills are taught and learned, particularly in language education. Over the past decade, educational technology research has shifted from examining access and delivery toward understanding how instructional design can support higher-order cognitive and metacognitive processes (Mayer, 2014; Bond et al., 2020). Within this landscape, academic speaking has emerged as a critical competence for university students, functioning not only as a communicative skill but also as a mechanism for knowledge construction, argumentation, and disciplinary participation (Zhu & Carless, 2018; Goh & Liu, 2023). Despite its importance, academic speaking instruction in higher education remains largely performance-oriented, often emphasizing fluency and linguistic accuracy while paying limited attention to the metacognitive processes that underpin effective spoken communication (Baines et al., 2025).

From a cognitive and educational psychology perspective, successful academic speaking requires learners to actively regulate their thinking before, during, and after oral performance. Metacognitive regulation, encompassing planning, monitoring, and evaluation, has been identified as a key factor influencing learners' ability to organize ideas, adapt explanations to audiences, and reflect on communicative effectiveness (Flavell, 1979; Magno, 2010; Zimmerman, 2002). Empirical studies in language education consistently demonstrate that learners with higher levels of metacognitive regulation exhibit stronger speaking performance and greater capacity for self-directed improvement (Goh & Liu, 2023; Sabnani & Goh, 2022). However, in many higher education contexts, metacognitive instruction remains implicit or incidental rather than systematically embedded within speaking pedagogy (Zhang et al., 2021).

Concurrently, the rise of digital-native learners has intensified calls for instructional designs that align with learners' cognitive characteristics and learning habits. Microlearning has gained prominence as a pedagogical approach that delivers content in short, focused segments designed to reduce cognitive overload and support active

processing (Hug & Friesen, 2007; Susilana et al., 2022). Research in educational technology suggests that microlearning is particularly effective when combined with reflective and generative learning activities, rather than being used solely for content transmission (Noetel et al., 2022). Grounded in the Cognitive Theory of Multimedia Learning, segmented and multimodal instruction can facilitate deeper processing when learners are required to actively organize and integrate information (Mayer, 2014; Fiorella & Mayer, 2016). Nevertheless, studies also caution that microlearning, if poorly designed, risks fragmenting understanding and reinforcing surface learning (De Jong, 2010; Jahnke et al., 2020).

One instructional strategy with strong potential to complement microlearning is the Feynman technique, an explain-to-learn approach that requires learners to articulate concepts in their own words and identify gaps in understanding through explanation (Reyes et al., 2021; Paterson, 2022). Rooted in principles of generative learning and metacognitive awareness, the Feynman technique has been shown to promote deeper conceptual understanding, reflective thinking, and communicative clarity across disciplines (Goldhawk, 2024; Fiorella & Mayer, 2016). In language education, explaining concepts aloud aligns closely with the cognitive demands of academic speaking, as learners must plan explanations, monitor clarity, and evaluate audience comprehension in real time (Goh & Liu, 2023). Despite these theoretical synergies, empirical research integrating microlearning and the Feynman technique to support metacognitive regulation in academic speaking remains limited, particularly in higher education contexts.

Within the Indonesian higher education landscape, recent policy orientations further underscore the relevance of this issue. The Kampus Berdampak paradigm emphasizes outcome- and impact-based learning, positioning communication competence as a strategic graduate attribute for knowledge dissemination and societal engagement (Yulianto et al., 2025). Similarly, the framework of Pembelajaran Mendalam highlights communication and metacognitive reflection as core dimensions of graduate profiles, calling for instructional practices that move beyond rote learning toward meaningful knowledge construction (Kementerian Pendidikan Dasar dan Menengah Republik

Indonesia, 2025). These policy directions amplify the need for instructional designs that explicitly integrate technology, metacognition, and academic communication.

Against this backdrop, the present study adopts a design-based research approach to examine the integration of microlearning and the Feynman technique as a means of fostering metacognitive regulation in academic speaking within higher education. By iteratively designing, implementing, and evaluating a technology-mediated instructional intervention, this study seeks to contribute empirical evidence on how explain-to-learn principles embedded in microlearning environments can support deeper learning processes in academic speaking. Specifically, this study addresses the following research questions 1) to what extent does the integration of microlearning and the Feynman technique influence students' metacognitive regulation (planning, monitoring, and evaluation) in academic speaking tasks? 2) How does instructional intervention affect students' academic speaking performance across different dimensions (idea organization, clarity of explanation, strategic delivery, and responsiveness)? 3) What are students' perceptions regarding the usefulness, cognitive engagement, and instructional clarity of the integrated microlearning and explain-to-learn approach?

Literature Review

Academic Speaking and Metacognitive Regulation in Higher Education

In higher education, academic speaking is no longer conceived merely as an oral extension of linguistic knowledge but as a complex cognitive activity through which learners externalize, negotiate, and reconstruct disciplinary understanding (Zhu & Carless, 2018; Mercer, 2018). Speaking in academic contexts requires students to do more than "talk fluently"; they must plan ideas, structure arguments, anticipate audiences, and evaluate the effectiveness of their explanations in real time (Goh & Liu, 2023). In this sense, academic speaking can be likened to thinking aloud under cognitive pressure—verbalizing thought while simultaneously monitoring its coherence and relevance.

From the perspective of educational psychology, this process is inseparable from metacognitive regulation. Metacognitive regulation refers to learners' ability to plan,

monitor, and evaluate their cognitive activities while performing a task (Flavell, 1979; Magno, 2010; Schraw, 1998). In speaking tasks, these processes manifest concretely: for example, when a student outlines key points before a presentation (planning), adjusts explanations in response to audience cues (monitoring), or reflects on the clarity of arguments after speaking (evaluation). Empirical studies consistently indicate that learners with stronger metacognitive regulation demonstrate higher-quality oral performance and greater autonomy in improving their speaking skills (Sabnani & Goh, 2022; Zhang et al., 2021; Teng & Zhang, 2016).

Despite this evidence, academic speaking instruction in universities often remains product-focused, emphasizing final performance outcomes such as fluency, pronunciation, or lexical accuracy while marginalizing the cognitive and metacognitive processes that generate those outcomes (Baines et al., 2025; Goh, 2018). This situation mirrors what has been described in other educational domains as "teaching the performance without teaching the thinking behind it" (Magno, 2010, p. 42). As a result, students may appear competent in rehearsed situations but struggle to transfer speaking skills to new academic contexts.

Microlearning as Instructional Design for Deep Processing

In response to the learning preferences of digital-native students, microlearning has gained prominence as an instructional design approach that delivers content in short, focused, and modular units (Hug & Friesen, 2007; Susilana et al., 2022). Rather than functioning as a simplified version of traditional instruction, microlearning is intended to reduce cognitive overload and support active processing by segmenting complex information into manageable learning episodes (Mayer, 2014; Fiorella & Mayer, 2016). One may think of microlearning as intellectual scaffolding in small steps, where each step is designed to be cognitively digestible but conceptually connected to a larger structure.

Research in educational technology suggests that microlearning can enhance engagement and retention, particularly when learning objectives are clearly defined and aligned with reflective activities (Susilana et al., 2022; Noetel et al., 2022). However, scholars have also cautioned against viewing microlearning as a panacea. When used merely to transmit fragmented information, it risks promoting surface learning and conceptual

disconnection, much like assembling puzzle pieces without a picture of the whole (De Jong, 2010; Jahnke et al., 2020). Meta-analytic evidence further indicates that the effectiveness of multimedia and microlearning designs depends heavily on how learners are prompted to actively organize and integrate information, rather than passively consume content (Noetel et al., 2022; Fiorella & Mayer, 2016).

Within the context of language education, microlearning has been applied primarily to vocabulary acquisition, grammar practice, and receptive skills (Susilana et al., 2022; Kasneci et al., 2023). Its application to productive skills such as academic speaking remains comparatively underexplored, particularly with respect to fostering metacognitive regulation rather than procedural repetition (Goh & Liu, 2023). This gap suggests that microlearning requires complementary pedagogical strategies that explicitly engage learners in reflective and generative processes.

The Feynman Technique as an Explain-to-Learn Strategy

One instructional strategy that aligns strongly with metacognitive and generative learning principles is the Feynman technique. Rooted in the idea that "what cannot be explained simply is not yet understood," the Feynman technique requires learners to explain concepts in their own words, identify gaps in understanding, and iteratively refine explanations (Reyes et al., 2021; Paterson, 2022). In educational terms, it operationalizes the principle of explain-to-learn, whereby explanation is not a by-product of learning but its driving mechanism (Paterson, 2022; Chi & Wylie, 2014).

The explanatory process inherent in the Feynman technique closely mirrors the cognitive demands of academic speaking. Explaining a concept orally requires learners to select essential information, organize it coherently, and adapt it to an imagined or real audience, all while monitoring clarity and accuracy (Goh & Liu, 2023). In this sense, the Feynman technique can be seen as a metacognitive mirror, forcing learners to confront what they truly understand versus what they merely recognize. Empirical studies across disciplines have shown that explain-to-learn strategies enhance conceptual understanding, critical thinking, and self-regulated learning (Goldhawk, 2024; Reyes et al., 2021; Fiorella & Mayer, 2016).

Nevertheless, research integrating the Feynman technique into technology-mediated environments remains limited, particularly in higher education language instruction. Most existing studies examine explanation as a classroom-based or peer-interaction activity, leaving open questions about how digital instructional designs can systematically embed explain-to-learn cycles and support metacognitive regulation over time.

Theoretical Framework

This study is grounded in two complementary theoretical perspectives: the Cognitive Theory of Multimedia Learning (CTML) and self-regulated learning (SRL) theory. These frameworks collectively inform the instructional design and provide a lens for interpreting how the integration of microlearning and the Feynman technique fosters metacognitive regulation in academic speaking.

Cognitive Theory of Multimedia Learning (CTML)

CTML posits that meaningful learning occurs when learners engage in active cognitive processing—selecting relevant information, organizing it into coherent mental representations, and integrating it with prior knowledge (Mayer, 2014). The theory emphasizes that instructional design should manage cognitive load through segmentation and signaling while promoting generative processing through active learning strategies. Table 1 presents the three core cognitive processes in CTML and how the instructional intervention operationalizes each process.

CTML Process	Description	Instructional Implementation
Selecting	Attending to relevant information	Microlearning segments focus attention on specific concepts; concise multimedia input reduces extraneous processing
Organizing	Structuring information into coherent representations	Feynman technique requires learners to organize ideas logically when explaining concepts in own words
Integrating	Connecting new information with prior knowledge	Explanation gaps identified during Feynman activity prompt connection-making and knowledge consolidation

Note. CTML = Cognitive Theory of Multimedia Learning

As illustrated in Table 1, the Feynman technique operationalizes all three cognitive processes of CTML: learners must select essential information when preparing

explanations (selecting), organize ideas coherently when articulating concepts (organizing), and connect new understanding with prior knowledge when identifying and addressing gaps (integrating). This alignment between instructional strategy and cognitive processing theory provides the theoretical foundation for hypothesizing that explain-to-learn activities embedded in microlearning environments will foster deeper learning outcomes.

Self-Regulated Learning (SRL) Theory

SRL theory complements CTML by foregrounding the metacognitive processes that govern learning (Zimmerman, 2002; Schraw, 1998). According to this perspective, effective learners actively plan their approach to tasks, monitor their understanding and performance during execution, and evaluate outcomes to inform future learning. These regulatory processes are particularly critical for complex skills like academic speaking, where real-time cognitive demands require ongoing self-management.

The integration of microlearning and the Feynman technique in this study was designed to systematically scaffold these metacognitive processes through structured pre-task, during-task, and post-task activities. Planning is supported through pre-speaking prompts that encourage learners to outline key points and anticipate audience needs. Monitoring is activated during the explanation phase as learners must attend to clarity and coherence in real time. Evaluation is facilitated through post-speaking reflection questions that prompt learners to assess their performance and identify areas for improvement. This tripartite structure aligns with the cyclical nature of self-regulated learning and provides multiple opportunities for metacognitive engagement throughout the learning process.

Integration of CTML and SRL in Research Design

The theoretical integration of CTML and SRL provides a comprehensive framework for understanding how the instructional intervention operates. CTML explains the cognitive mechanisms through which explain-to-learn activities promote learning (generative processing), while SRL theory illuminates the regulatory mechanisms that govern how learners manage their own learning processes. Together, these theories predict that the integration of microlearning (structural support for cognitive processing) and the Feynman technique (generative strategy activating regulatory processes) will produce

synergistic effects on both metacognitive regulation and academic speaking performance. Table 2 maps each research question to the theoretical constructs and expected outcomes.

RQ	Theoretical Construct	CTML Process	Expected Outcome
RQ1	Metacognitive regulation (SRL)	All three processes	Increased planning, monitoring, evaluation
RQ2	Academic speaking performance	Organizing, Integrating	Improved organization and clarity
RQ3	Learner perceptions	Selecting, Organizing	High usefulness and engagement ratings

Note. RQ = Research Question; SRL = Self-Regulated Learning; CTML = Cognitive Theory of Multimedia Learning

Research Method

Participants

The study involved 96 undergraduate students (Age $M = 20.4$ years, $SD = 1.2$; Females $n = 62$, 64.6%; Males $n = 34$, 35.4%) enrolled in a Language Education program at a public university in Indonesia. All participants were native speakers of Indonesian and had completed at least one year of formal academic speaking coursework prior to the study. The cohort can therefore be described as linguistically homogeneous in terms of first language background, yet heterogeneous in academic speaking proficiency and metacognitive awareness, a condition deemed suitable for design-based research focusing on instructional processes rather than population generalization (McKenney & Reeves, 2019).

Participants were recruited through intact classes to preserve ecological validity, as the intervention was embedded within regular instructional activities (i.e., authentic classroom conditions rather than laboratory manipulation). Participation was voluntary, and informed consent was obtained prior to data collection. Ethical considerations followed institutional research guidelines and principles of educational research involving adult learners, including anonymity, non-evaluative participation, and the absence of grading incentives (Creswell & Plano Clark, 2018).

Research Design

This study adopted a design-based research (DBR) approach to examine how the integration of microlearning and the Feynman technique fosters metacognitive regulation

in academic speaking. DBR was selected because it allows for the systematic investigation of learning processes while simultaneously refining instructional design through iterative cycles of implementation and analysis (Design-Based Research Collective, 2003; McKenney & Reeves, 2019; Wang & Hannafin, 2005).

The research design unfolded across three iterative phases over eight weeks:

1. Analysis and exploration (Week 1): Focusing on learners' needs, difficulties in academic speaking, and baseline assessment of metacognitive regulation.
2. Design and construction (Weeks 2-3): Involving the development of a web-based microlearning environment embedding explain-to-learn (Feynman-based) activities, followed by pilot testing.
3. Implementation and evaluation (Weeks 4-8): Where the instructional design was enacted across multiple cycles, with ongoing refinement based on formative feedback, culminating in post-intervention assessment.

Rather than testing a single causal hypothesis in isolation, the study sought to identify design principles and learning mechanisms (i.e., how and why the intervention works), a defining characteristic of DBR (Reeves, 2006; McKenney & Reeves, 2019).

Instructional Intervention

The intervention consisted of a web-based microlearning platform designed to support academic speaking through segmented learning units (bite-sized modules). The platform was developed using Moodle with custom plugins to support audio recording and self-reflection features. Each module focused on a specific speaking function (e.g., explaining concepts, structuring arguments, responding to questions) and followed a consistent instructional sequence:

1. Concept exposure (5-7 minutes): Students engaged with short multimedia input including video explanations, infographics, and annotated examples.
2. Explain-to-learn task (10-15 minutes): Students explained the concept orally using their own words, recording their explanation via the platform (Feynman technique).

3. Metacognitive reflection (5 minutes): Students responded to structured prompts encouraging planning (before speaking), monitoring (during explanation), and evaluation (after performance).

This structure was grounded in the Cognitive Theory of Multimedia Learning, particularly the principles of segmentation, limited capacity, and active processing (Mayer, 2014). The Feynman technique functioned as a generative strategy, transforming microlearning from a content-delivery mechanism into a reflective learning process (Reyes et al., 2021; Fiorella & Mayer, 2016).

Instruments and Data Collection

Data were collected using three complementary instruments to capture both outcomes and processes:

Academic Speaking Performance Test

Speaking performance was assessed through task-based oral presentations aligned with course objectives. The assessment rubric measured four dimensions: (a) idea organization, (b) clarity of explanation, (c) strategic delivery, and (d) responsiveness to audience cues. Each dimension was rated on a 5-point scale (1 = poor, 5 = excellent) by two independent raters. The rubric was adapted from established frameworks in academic speaking assessment (Zhu & Carless, 2018; Goh & Liu, 2023). Inter-rater reliability was satisfactory (Cohen's κ = .84 for pre-test, .87 for post-test).

Metacognitive Regulation Questionnaire

Metacognitive regulation was measured using a self-report questionnaire focusing on three dimensions: planning (5 items), monitoring (5 items), and evaluation (5 items). Items were adapted from validated metacognitive instruments and contextualized for academic speaking tasks (Magno, 2010; Sabnani & Goh, 2022). Example items include: "Before speaking, I plan how to organize my ideas" (planning), "While explaining, I check whether my audience understands" (monitoring), and "After speaking, I reflect on what I could improve" (evaluation). Responses were recorded on a 5-point Likert scale (1 = strongly disagree, 5 = strongly agree) to capture sufficient variance while maintaining cognitive accessibility for respondents.

Learner Perception Survey

To capture learners' experiences with the instructional design, a perception survey examined perceived usefulness, cognitive engagement, and clarity of instructional structure using 8 items (e.g., "The short learning units helped me focus on explaining ideas more clearly," "Explaining concepts in my own words improved my understanding"). This instrument provided qualitative support for interpreting quantitative trends (Creswell & Plano Clark, 2018).

Validity and Reliability

Content validity of the instruments was ensured through expert review by two scholars in language education and educational technology. Pilot testing was conducted with a small group of students ($n = 15$) sharing similar characteristics to the main participants, allowing for item refinement and clarity adjustments.

Internal consistency reliability was examined using Cronbach's alpha. All retained subscales exceeded the commonly accepted threshold of .70, indicating satisfactory reliability for exploratory and design-based research purposes (Tabachnick & Fidell, 2001; Nunnally & Bernstein, 1994). Construct validity was further supported through exploratory factor analysis, confirming the alignment between theoretical constructs and observed item groupings. Table 3 presents the reliability coefficients for each instrument.

Instrument	Dimension	Cronbach's α	Items
Metacognitive Regulation Questionnaire	Planning	.84	5
Metacognitive Regulation Questionnaire	Monitoring	.78	5
Metacognitive Regulation Questionnaire	Evaluation	.86	5
Academic Speaking Performance Test	Overall	.87	4
Learner Perception Survey	Overall	.82	8

Data Analysis

Quantitative data were analyzed using descriptive statistics and paired-sample comparisons to examine changes in metacognitive regulation and speaking performance across the intervention period. Effect sizes were calculated using Cohen's d for paired

samples and partial eta squared (η^2) for multivariate analyses. Qualitative data from learner perceptions and reflective responses were analyzed thematically following Braun and Clarke's (2006) six-phase approach to identify recurring patterns related to metacognitive engagement and instructional design features. Integration of quantitative and qualitative findings followed a convergent mixed-methods logic, enabling triangulation and richer interpretation of results (Creswell & Plano Clark, 2018).

Results

To examine the effectiveness of the instructional design integrating microlearning and the Feynman technique, the analytical procedure focused on testing inferences regarding the relationship between metacognitive regulation and academic speaking performance following the intervention. Prior to hypothesis testing, assumptions of normality (Shapiro-Wilk test), homogeneity of variance (Levene's test), and intercorrelation among variables were examined and met at acceptable levels for multivariate analysis (Tabachnick & Fidell, 2001).

Descriptive Statistics

Table 4 presents the means and standard deviations for all measures at pre-test and post-test. Results indicate improvement across all variables, with the largest gains observed in evaluation and planning dimensions of metacognitive regulation.

Variable	Pre-test M (SD)	Post-test M (SD)	MD
Planning	3.12 (0.78)	4.17 (0.62)	1.05
Monitoring	3.45 (0.71)	4.02 (0.68)	0.57
Evaluation	3.08 (0.82)	4.26 (0.59)	1.18
Speaking Performance	2.89 (0.74)	3.94 (0.65)	1.05

Note. MD = Mean Difference

RQ1: Effects on Metacognitive Regulation

To address Research Question 1 regarding the influence of the instructional intervention on students' metacognitive regulation, a repeated-measures MANOVA was conducted using the three metacognitive dimensions (planning, monitoring, evaluation) as dependent variables. Results revealed a statistically significant multivariate effect of the intervention on metacognitive regulation, Wilks' $\Lambda = .41$, $F(3, 93) = 44.62$, $p < .001$, partial $\eta^2 = .59$. According to conventional benchmarks (Cohen, 1988), this value reflects

a large effect size, indicating that the instructional design accounted for a substantial proportion of variance in students' metacognitive regulation.

Univariate follow-up analyses indicated differential effects across dimensions. Planning showed a strong effect, $F(1, 95) = 61.84$, $p < .001$, partial $\eta^2 = .39$, followed by evaluation, $F(1, 95) = 68.12$, $p < .001$, partial $\eta^2 = .42$. Monitoring also demonstrated a statistically significant improvement, though with a comparatively moderate effect size, $F(1, 95) = 27.46$, $p < .001$, partial $\eta^2 = .22$. These results suggest that the intervention was particularly effective in fostering pre-task planning and post-task reflection (i.e., before and after speaking), while in-task monitoring developed to a lesser yet still meaningful extent. Table 5 presents the complete univariate ANOVA results.

Dimension	F(1, 95)	p-value	Partial η^2	Effect Size
Planning	61.84	< .001	.39	Large
Monitoring	27.46	< .001	.22	Medium
Evaluation	68.12	< .001	.42	Large

Paired-sample comparisons further confirmed these findings. Planning demonstrated a significant increase from pre-test ($M = 3.12$, $SD = 0.78$) to post-test ($M = 4.17$, $SD = 0.62$), $t(95) = 8.74$, $p < .001$, Cohen's $d = 0.89$. Evaluation showed the largest gain, increasing from $M = 3.08$ ($SD = 0.82$) to $M = 4.26$ ($SD = 0.59$), $t(95) = 9.21$, $p < .001$, Cohen's $d = 0.94$. Monitoring also improved significantly, though with a smaller effect size, from $M = 3.45$ ($SD = 0.71$) to $M = 4.02$ ($SD = 0.68$), $t(95) = 5.63$, $p < .001$, Cohen's $d = 0.58$. These findings provide strong support for the hypothesis that integrating microlearning with the Feynman technique significantly enhances students' metacognitive regulation in academic speaking, with the strongest effects observed in planning and evaluation dimensions.

RQ2: Effects on Academic Speaking Performance

Research Question 2 examined how the instructional intervention affected students' academic speaking performance across different dimensions. A paired-sample t-test comparing pre- and post-intervention speaking scores revealed a statistically significant increase in overall speaking performance, $t(95) = 9.87$, $p < .001$, with a large effect size (Cohen's $d = 1.01$).

Further inspection of performance components revealed that gains were most pronounced in idea organization and clarity of explanation. Idea organization demonstrated the largest improvement, increasing from $M = 2.76$ ($SD = 0.81$) to $M = 3.97$ ($SD = 0.68$), $MD = 1.21$, Cohen's $d = 1.15$. Clarity of explanation also showed substantial gains, from $M = 2.84$ ($SD = 0.79$) to $M = 3.92$ ($SD = 0.71$), $MD = 1.08$, Cohen's $d = 0.98$. Strategic delivery improved from $M = 2.95$ ($SD = 0.76$) to $M = 3.79$ ($SD = 0.72$), $MD = 0.84$, Cohen's $d = 0.87$. Responsiveness to audience cues showed the smallest but still significant improvement, from $M = 3.02$ ($SD = 0.74$) to $M = 3.81$ ($SD = 0.69$), $MD = 0.79$, Cohen's $d = 0.82$.

Table 6 presents the complete paired-sample t-test results for all speaking performance dimensions. The pattern of results suggests that students became more adept at structuring and articulating academic explanations rather than merely improving surface fluency. This pattern aligns closely with the pedagogical emphasis on explain-to-learn activities embedded in the microlearning design and supports the theoretical prediction that generative learning strategies would enhance the organizational and explanatory dimensions of speaking performance.

Dimension	t (95)	p-value	Cohen's d	Interpretation
Idea Organization	10.12	< .001	1.15	Large
Clarity of Explanation	9.56	< .001	0.98	Large
Strategic Delivery	8.43	< .001	0.87	Large
Responsiveness	7.89	< .001	0.82	Large
Overall Performance	9.87	< .001	1.01	Large

Correlation analysis further illuminated the relationship between metacognitive regulation and speaking performance. Academic speaking performance showed consistent and statistically significant correlations with all three metacognitive dimensions: planning ($r = .68$, $p < .01$), monitoring ($r = .54$, $p < .01$), and evaluation ($r = .71$, $p < .01$). The strongest correlation was observed between evaluation and speaking performance, suggesting that students' capacity for post-task reflection may be particularly important for speaking development. These results support the theoretical proposition that metacognitive regulation functions as a mediating mechanism linking instructional design to speaking performance outcomes.

RQ3: Student Perceptions of the Instructional Approach

Research Question 3 investigated students' perceptions regarding the usefulness, cognitive engagement, and instructional clarity of the integrated microlearning and explain-to-learn approach. Both quantitative and qualitative data were analyzed to provide a comprehensive understanding of learner experiences.

Quantitative results from the perception survey indicated positive student responses across all measured dimensions. Perceived usefulness received a mean rating of $M = 4.23$ ($SD = 0.71$) on a 5-point scale, with 78% of students rating this dimension at 4 or above. Cognitive engagement showed similarly positive ratings ($M = 4.15$, $SD = 0.76$), with 74% of students providing ratings of 4 or higher. Instructional clarity was also well-received ($M = 4.08$, $SD = 0.79$), suggesting that students found the structure and sequencing of learning activities easy to follow.

Qualitative analysis of open-ended responses revealed three overarching themes. First, students consistently reported that the segmented structure of microlearning modules reduced cognitive overwhelm and enabled focused attention on specific speaking tasks. Representative comments included: "The short videos helped me understand one thing at a time without feeling overwhelmed" and "Breaking down the learning into small parts made it easier to focus." This theme aligns with the CTML principle of segmentation and supports the design decision to deliver content in manageable units.

Second, students valued the explain-to-learn activities as opportunities to consolidate understanding and identify knowledge gaps. Comments in this theme included: "When I had to explain the concept myself, I realized what I didn't really understand," "Recording my explanation made me think more carefully about my words," and "Teaching the concept to myself showed me where I was confused." These responses provide qualitative evidence that the Feynman technique activated the generative processing mechanisms posited by CTML, particularly the processes of organizing and integrating information.

Third, students appreciated the metacognitive prompts as tools for developing self-awareness and self-regulation. Representative comments included: "The reflection

questions after each task made me think about how I can improve next time," "I started planning my explanations more carefully because of the prompts," and "The questions helped me notice when my explanation wasn't clear." These findings suggest that the structured reflection component successfully scaffolded the development of metacognitive regulation, supporting the theoretical integration of CTML and SRL in the instructional design.

Table 7 summarizes the quantitative perception data, while Table 8 presents illustrative qualitative responses organized by theme.

Dimension	M	SD	% Rating 4-5
Perceived Usefulness	4.23	0.71	78%
Cognitive Engagement	4.15	0.76	74%
Instructional Clarity	4.08	0.79	71%
Theme	Illustrative Quote		
Segmentation Reduces Overwhelm	"The short videos helped me understand one thing at a time without feeling overwhelmed."		
Explain-to-Learn Consolidates Understanding	"When I had to explain the concept myself, I realized what I didn't really understand."		
Metacognitive Prompts Develop Self-Awareness	"The reflection questions after each task made me think about how I can improve next time."		

Discussion

This study investigated how the integration of microlearning and the Feynman technique fosters metacognitive regulation as a mediating mechanism for academic speaking development in higher education. Grounded in the Cognitive Theory of Multimedia Learning and self-regulated learning theory, the research provides empirical evidence for the pedagogical value of technology-mediated explain-to-learn approaches in language education. The discussion addresses each research question in turn, situating findings within the theoretical framework and prior research.

Interpretation of Findings on Metacognitive Regulation (RQ1)

The finding that the instructional intervention produced significant improvements in all dimensions of metacognitive regulation supports the theoretical prediction that explain-to-learn activities embedded in microlearning environments can systematically activate regulatory processes. The large multivariate effect size (partial $\eta^2 = .59$) indicates that the instructional design accounted for a substantial proportion of variance in students' metacognitive regulation, suggesting that the integration of microlearning and the Feynman technique was effective in fostering the planning, monitoring, and evaluation processes essential for self-regulated learning.

Notably, the strongest effects were observed in planning and evaluation, while monitoring exhibited comparatively moderate gains. This asymmetry is theoretically meaningful and aligns with prior research on metacognitive instruction. Planning and evaluation are temporally distanced from the immediacy of speaking performance and thus more amenable to instructional scaffolding through prompts, checklists, and reflective tasks. Monitoring, by contrast, unfolds in real time under cognitive load, resembling what might be described as thinking on one's feet—regulating cognition while simultaneously producing language. The relative resistance of monitoring to instructional change has been reported in previous studies and underscores the challenge of fostering in-the-moment metacognitive control through short-term interventions (Zhang et al., 2021; Sabnani & Goh, 2022; Teng & Zhang, 2016).

From a CTML perspective, the differential effects across metacognitive dimensions can be understood through the lens of cognitive load management. Planning and evaluation occur outside the primary speaking task, allowing learners to devote cognitive resources to regulatory processes without the simultaneous demands of language production. Monitoring, however, must compete with the cognitive resources required for real-time speaking, making it more difficult to develop through instructional scaffolding alone. Future designs might incorporate additional supports for in-task monitoring, such as real-time feedback or structured pauses during speaking tasks.

Interpretation of Findings on Academic Speaking Performance (RQ2)

The significant improvement in academic speaking performance (Cohen's $d = 1.01$) provides evidence that the instructional intervention successfully enhanced students' ability to communicate academic content orally. The pattern of gains—most pronounced in idea organization and clarity of explanation—aligns closely with the explain-to-learn logic underpinning the Feynman technique. By requiring learners to externalize understanding through explanation, the intervention activated the generative processing mechanisms posited by CTML, particularly the processes of selecting essential information, organizing ideas coherently, and integrating new understanding with prior knowledge.

The relatively smaller gains in responsiveness to audience cues may reflect the nature of the intervention, which focused primarily on individual explanation tasks rather than interactive communication. While the Feynman technique emphasizes clarity and coherence in explanation, it does not inherently require learners to adapt their communication in response to real-time audience feedback. Future iterations of the instructional design might incorporate more interactive elements to foster this dimension of speaking competence.

The strong correlations observed between metacognitive regulation dimensions and speaking performance (ranging from $r = .54$ to $r = .71$) support the theoretical proposition that metacognitive regulation functions as a mediating mechanism linking instructional design to learning outcomes. Students who demonstrated stronger regulatory capabilities also showed greater improvements in speaking performance, suggesting that the development of metacognitive skills was not merely an incidental outcome but a critical pathway through which the intervention produced its effects.

Interpretation of Findings on Student Perceptions (RQ3)

The positive student perceptions reported across all dimensions—perceived usefulness, cognitive engagement, and instructional clarity—provide important qualitative support for the quantitative findings on learning outcomes. The high ratings for perceived usefulness ($M = 4.23$) suggest that students recognized the value of the integrated

approach for their learning, which may have contributed to their motivation and engagement throughout the intervention.

The three themes emerging from qualitative analysis—segmentation reducing cognitive overwhelm, explain-to-learn consolidating understanding, and metacognitive prompts developing self-awareness—provide rich evidence that students experienced the instructional design as intended. The alignment between student reports and theoretical predictions strengthens confidence in the interpretation that the intervention operated through the hypothesized mechanisms of generative processing and metacognitive regulation.

Comparison with Prior Research

The effect sizes observed in this study (partial $\eta^2 = .59$ for metacognitive regulation, Cohen's $d = 1.01$ for speaking performance) are comparable to or exceed those reported in previous studies examining metacognitive instruction in language learning. For example, Teng and Zhang (2016) reported Cohen's $d = 0.72$ for the effect of metacognitive instruction on writing performance, while Zhang et al. (2021) found moderate effects ($r = .45$) for the relationship between metacognitive awareness and speaking performance. The comparatively stronger effects in this study may be attributed to the systematic integration of metacognitive scaffolding within a technology-mediated environment, as opposed to more generic metacognitive instruction.

The findings also align with research on generative learning strategies, which consistently demonstrates that activities requiring learners to actively explain concepts produce deeper learning than passive study (Fiorella & Mayer, 2016; Chi & Wylie, 2014). The integration of the Feynman technique within microlearning modules represents a novel application of these principles to academic speaking instruction, extending prior research on explain-to-learn strategies to a new domain and modality.

Limitations

Several limitations of this study should be acknowledged. First, the study relied on a single institutional context and a relatively modest sample size ($N = 96$), limiting the generalizability of the findings to diverse higher education settings. Replication studies

across different institutional contexts, cultural backgrounds, and language learning environments would strengthen the evidence base for the instructional approach.

Second, the use of self-report measures for metacognitive regulation, while common in the literature, may not fully capture learners' real-time regulatory behavior. Self-report instruments assess learners' perceptions of their regulatory processes rather than their actual cognitive activities during task performance. Future research could incorporate process-oriented data sources such as stimulated recall interviews, think-aloud protocols, or learning analytics (e.g., pause patterns, reformulations during speech) to triangulate self-reported regulation with behavioral measures (Winne & Jamieson-Noel, 2002).

Third, the eight-week intervention period, while sufficient to observe meaningful changes in metacognitive regulation and speaking performance, may not capture the full trajectory of skill development. Longitudinal studies examining sustained effects over longer periods (e.g., semester-long or year-long interventions) would provide valuable insights into the durability of learning gains and the potential for continued development with extended practice.

Fourth, the design-based nature of the study prioritizes explanatory insight and instructional refinement over broad causal generalization. While the iterative approach strengthens internal validity and produces actionable design principles, the findings should be interpreted as evidence of what is possible under specific conditions rather than universal claims about the effectiveness of the intervention across all contexts.

Fifth, the study did not include a control group receiving alternative instruction, limiting the ability to make strong causal claims about the unique contributions of the integrated approach compared to other instructional methods. Future research employing randomized controlled designs would strengthen causal inferences and provide more definitive evidence of the intervention's efficacy relative to baseline or alternative conditions.

Recommendations for Future Research

Based on the findings and limitations of this study, several directions for future research are recommended:

First, future studies should investigate the long-term sustainability of metacognitive regulation and speaking performance gains through longitudinal designs. Understanding whether the benefits observed in this eight-week intervention persist over time—and under what conditions—is critical for assessing the practical value of the instructional approach.

Second, research should explore the integration of process-oriented measures to complement self-report assessments of metacognitive regulation. Methods such as stimulated recall, discourse analysis of spoken output, and learning analytics could provide richer, more nuanced understanding of how regulatory processes unfold during academic speaking tasks.

Third, comparative studies examining the integrated microlearning and Feynman technique approach against alternative instructional methods would clarify the unique contributions of each component. Such research could address questions about whether the benefits observed are attributable to microlearning structure, explain-to-learn activities, their integration, or some combination thereof.

Fourth, research should investigate the applicability of the instructional approach across diverse learner populations, including different proficiency levels, academic disciplines, and cultural contexts. Understanding how the intervention functions across varied populations would inform adaptations needed for broader implementation.

Fifth, future designs might explore the incorporation of additional supports for in-task monitoring, which showed comparatively smaller gains in this study. Strategies such as real-time feedback, structured pauses during speaking, or peer interaction components could potentially enhance the development of real-time regulatory capabilities.

Practical Implications

The findings of this study carry several practical implications for educators, instructional designers, and policymakers in higher education language instruction:

For educators, the study demonstrates that metacognitive regulation can be systematically fostered through carefully designed instructional activities. Rather than assuming that metacognitive skills develop incidentally through language practice, instructors can explicitly scaffold planning, monitoring, and evaluation processes through structured prompts and reflective tasks. The specific sequence of concept exposure, explain-to-learn activity, and metacognitive reflection provides a replicable model for designing speaking instruction that targets regulatory processes.

For instructional designers, the study offers evidence-based design principles for technology-enhanced language learning environments. Key principles include: (1) segmenting content into manageable units to reduce cognitive overload, (2) embedding generative learning activities that require active explanation rather than passive consumption, (3) providing structured metacognitive prompts at key points in the learning process, and (4) integrating audio recording and reflection features that enable self-assessment and iterative improvement.

For policymakers, the findings support contemporary orientations emphasizing deep learning and communication competence in higher education. The Kampus Berdampak and Pembelajaran Mendalam frameworks in Indonesia, as well as similar initiatives globally, call for instructional practices that move beyond surface-level skill development toward meaningful knowledge construction and metacognitive reflection. This study provides empirical evidence that technology-mediated instructional designs can effectively support these policy goals when grounded in established learning theories and implemented with attention to cognitive and metacognitive processes.

For technology developers, the study highlights the importance of designing learning platforms that support generative processing and metacognitive regulation rather than merely delivering content. Features such as audio recording, structured reflection prompts, and progress tracking can transform microlearning environments from passive

content delivery systems into active learning tools that foster deeper engagement and self-regulated learning.

Conclusion

This study contributes to the growing body of research that conceptualizes learning as a dynamic, regulated process shaped by instructional design and learner agency. By demonstrating how microlearning integrated with the Feynman technique can foster metacognitive regulation as a mediating mechanism for academic speaking development, the study advances a process-oriented perspective that moves beyond achievement-focused paradigms.

The findings provide robust empirical support for the theoretical proposition that metacognitive regulation functions as a critical pathway linking explain-to-learn instructional design to speaking performance outcomes. The large effect sizes observed for both metacognitive regulation (partial $\eta^2 = .59$) and speaking performance (Cohen's $d = 1.01$) demonstrate that the integrated approach produced meaningful, educationally significant improvements in student learning.

The positive student perceptions reported across all dimensions—perceived usefulness, cognitive engagement, and instructional clarity—suggest that the intervention was not only effective but also well-received by learners. The alignment between quantitative outcomes and qualitative experiences strengthens confidence in the interpretation that the intervention operated through the hypothesized mechanisms of generative processing and metacognitive regulation.

Such a perspective invites further inquiry into how higher education language instruction can cultivate sustainable communicative competence, positioning academic speaking not merely as an assessed outcome but as a reflective practice that evolves through guided regulation and purposeful engagement. The design principles and theoretical insights generated through this research provide a foundation for continued innovation in technology-enhanced language education, with implications extending beyond the specific context of academic speaking to broader questions about how digital learning environments can support self-regulated learning across domains.

The authors declare that there is no conflict of interest in this work.

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